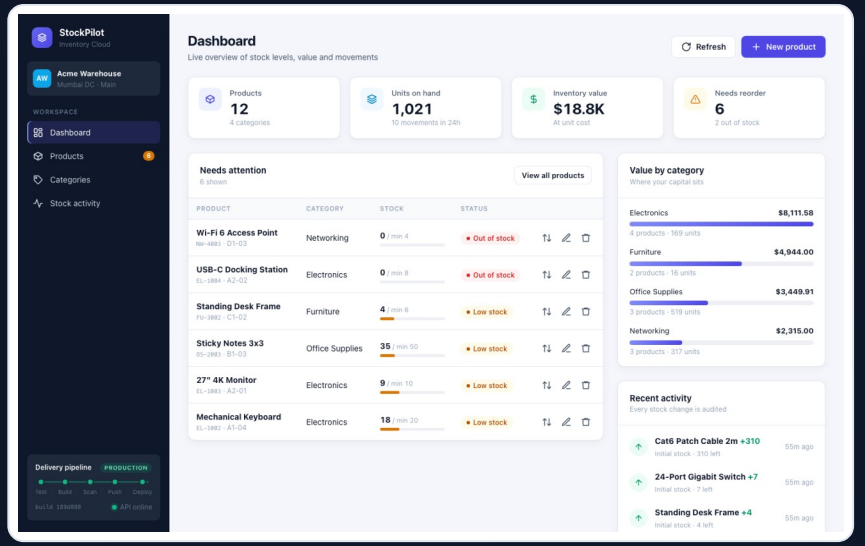
StockPilot

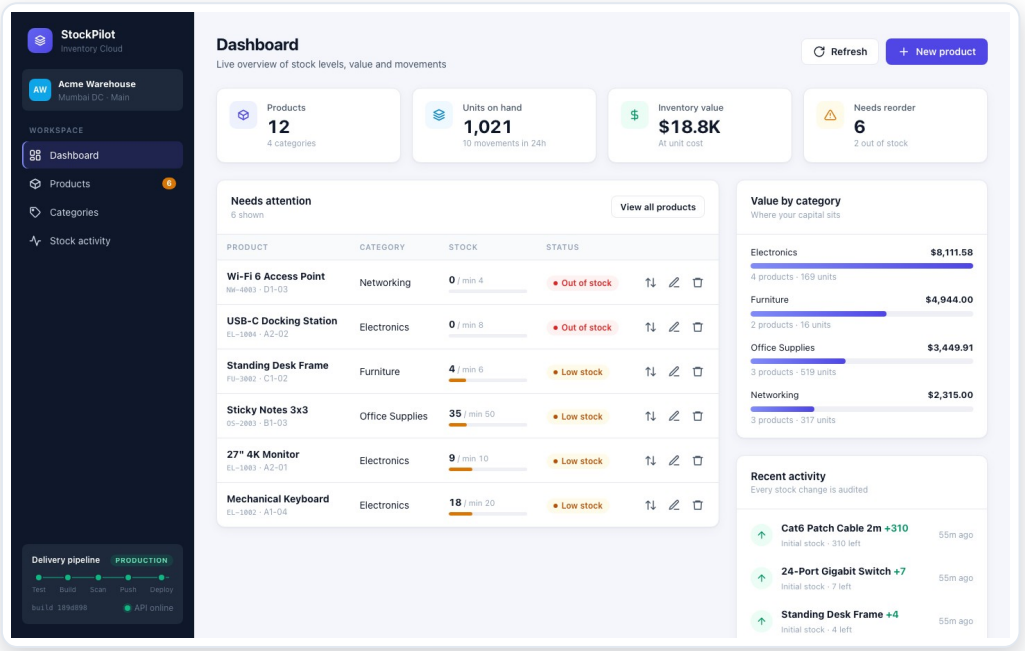
Inventory management, delivered from a Git commit to a monitored Kubernetes cluster on AWS



DevOps Final Capstone

Dhruv · October 2026

The application



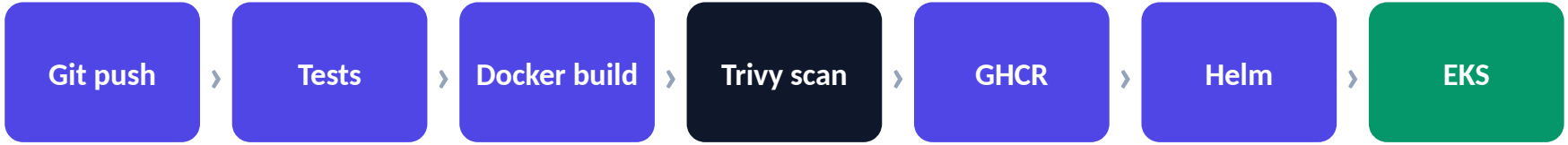
1. **React + Vite frontend**
   Responsive dashboard, served by Nginx

2. **FastAPI backend**
   14 REST endpoints + /health, /ready, /metrics

3. **PostgreSQL**
   SQLAlchemy models, Alembic migrations

4. **32 automated tests**
   24 pytest + 8 Vitest, run on every push

From commit to cluster



Commit SHA

is the image tag on every build

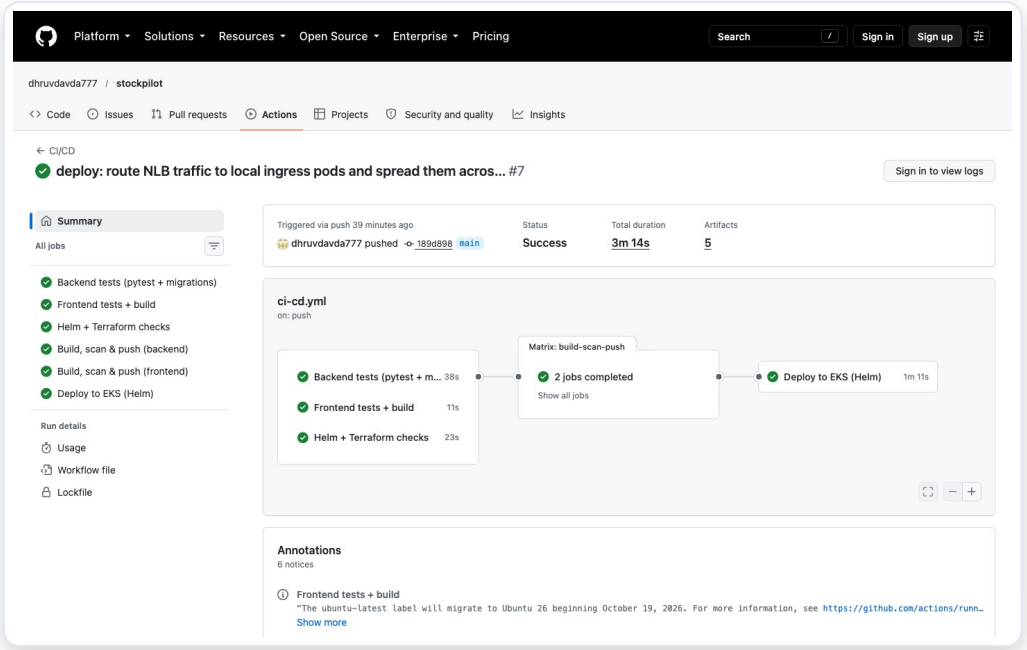
Quality gate

failed test or CVE = no release

Keyless

OIDC to AWS, no stored access keys

CI/CD and security



6 / 6 jobs

green, ending in the EKS deploy

0

HIGH/CRITICAL CVEs, both images

SHA-pinned

every third-party GitHub Action

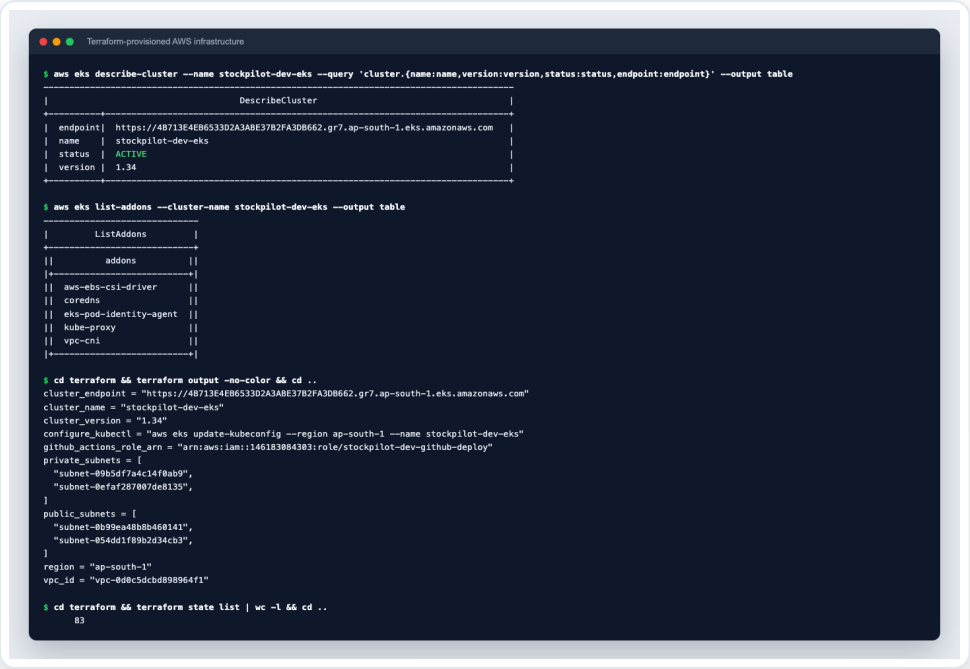
Infrastructure as code

1. **VPC across 2 AZs**
   Public subnets + NAT, private subnets for nodes

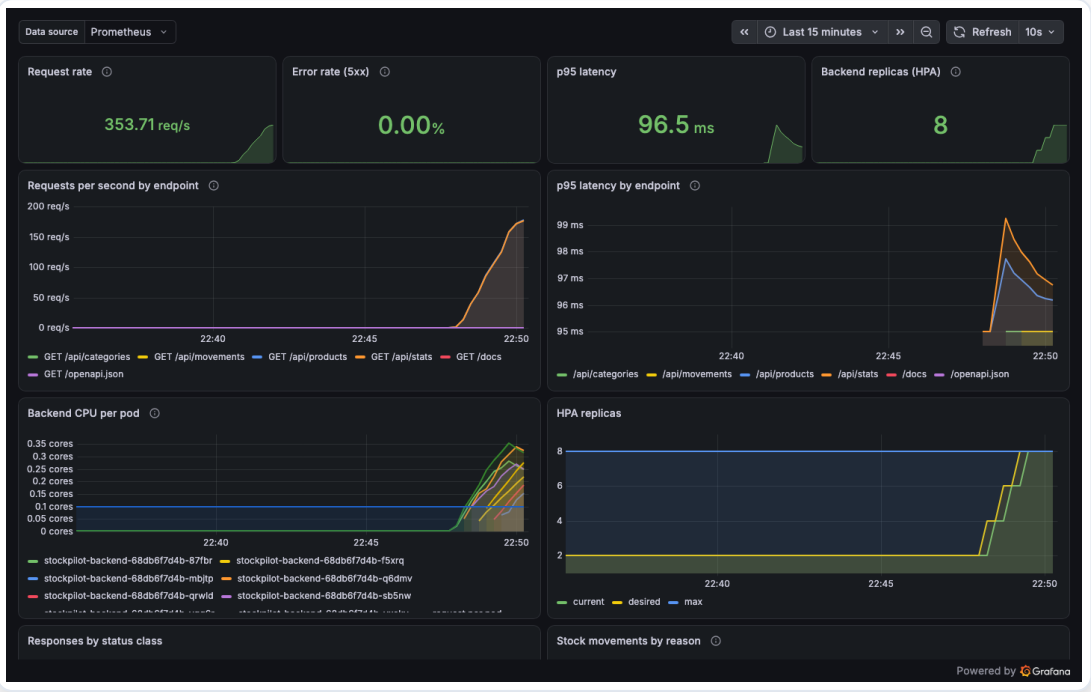
2. **EKS 1.34 + managed nodes**
   VPC-CNI, CoreDNS, EBS CSI via Pod Identity

3. **GitHub OIDC deploy role**
   Scoped to main and the production environment

4. **64 resources, one command**
   terraform plan, apply, and destroy when done



Autoscaling and monitoring



2 → 8 pods

HPA scaled the backend under load

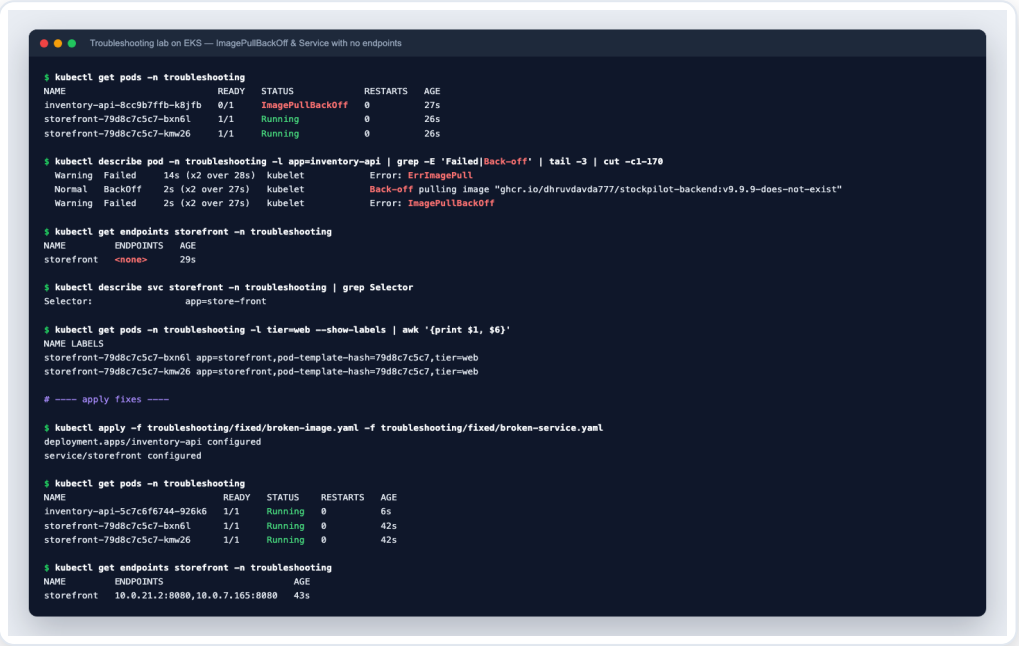
353 req/s

with 96 ms p95 latency

0.00%

errors, scraped by Prometheus

Troubleshooting lab



1. **ImagePullBackOff**
   Wrong image tag, found with describe pod
2. **Service with no endpoints**
   Selector didn't match the pod labels
3. **Probe, config and memory**
   0/1 Ready, CrashLoopBackOff, OOMKilled
4. **Database outage**
   Pods leave the Service, no restarts

Every step automated and observable

32	0	2 → 8	5
automated tests	critical CVEs	pods autoscaled	failure drills

github.com/dhruvdavda777/stockpilot